

Time Series

Applied Time Series Analysis **STA457 Lecture Notes**

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Time Series Analysis

Chapter 4: Linear Time Series Models

Invertibility of an MA Process

- 1 Let us consider an $MA(1)$ process:

$$X_t = \mu + W_t + \theta W_{t-1}. \quad (1)$$

- 2 We have already shown that this process is stationary even in the case that no restriction is put on the parameter θ . But, it is generally desirable to impose some restrictions on θ to ensure that the process satisfies a condition called *invertibility*.

- 3 To clarify the concept, let us take a look at two MA processes:

$$A : X_t = W_t + \theta W_{t-1} \quad (2)$$

$$B : X_t = W_t + \frac{1}{\theta} W_{t-1} \quad (3)$$

- 4 It can be shown that these two different processes have exactly the same ACF. Thus, we cannot identify a MA process uniquely from a given ACF.
- 5 Now, if we invert models A and B by expressing W_t in terms of $X_t, X_{t-1}, \dots$, we find by successive substitution that

$$A : W_t = X_t - \theta X_{t-1} + \theta^2 X_{t-2} - \dots \quad (4)$$

$$B : W_t = X_t - \frac{1}{\theta} X_{t-1} + \frac{1}{\theta^2} X_{t-2} - \dots \quad (5)$$

Conditions for Invertibility

- ① if $|\theta| < 1$, the series of coefficients of X_{t-j} for model A converges whereas that of B does not. Thus, model B cannot be inverted in this way.
- ② More generally, a process $\{X_t\}$ is said to be invertible if the random disturbance at time t , sometimes called the *innovation*, can be expressed as a convergent sum of present and past values of X_t in the form:

$$W_t = \sum_{j=0}^{\infty} \pi_j X_{t-j}, \quad (6)$$

where $\sum |\pi_j| < \infty$.

- ③ It turns out, an estimation procedure for a MA process, which involves estimating the residuals, will lead naturally to model A.
- ④ Thus, if $|\theta| < 1$, model A is said to be invertible whereas model B is not.
- ⑤ The imposition of invertibility condition ensures that there is a *unique* invertible first-order MA process for a given *ACF*.

Duality Between AR and MA Processes

- 1 In the previous slide, we saw that the first-order MA process could effectively be written in the form of an autoregressive model, possibly of infinite order, whose coefficients form a convergent sum.
- 2 It is also interesting that a similar property holds true for AR process. For instance, we showed earlier that an AR(1) process can be written as

$$X_t = W_t + \phi W_{t-1} + \phi^2 W_{t-2} + \dots \quad (7)$$

provided $|\phi| < 1$, so that the sum converges. But, this is just in form of an MA(∞) process.

- 3 The possibility that AR processes may be written in MA form, and vice versa, means that there is a *duality* between AR and MA processes, which is useful for a variety of purposes.

Conditions for Invertibility Using Backward Shift Operator

- 1 The invertibility condition for a MA process of any order is best expressed by using the backward shift operator, denoted by B , which is defined by

$$B^j X_t = X_{t-j} \quad \text{for all } j \quad (8)$$

- 2 Then,

$$X_t = (\beta_0 + \beta_1 B + \cdots + \beta_q B^q) W_t = \theta(B) W_t \quad (9)$$

where $\theta(B)$ is a polynomial of order q in B .

- 3 It can be shown that a MA(q) process is invertible if the roots of the equation

$$\theta(B) = \beta_0 + \beta_1 B + \cdots + \beta_q B^q = 0 \quad (10)$$

all lie outside of the unit circle, where we regard B as a complex variable and not as an operator. This means that the roots, which may be complex, have modulus greater than unity.

- 4 Also, note that

$$W_t = \frac{1}{\theta(B)} X_t \quad (11)$$

In the first-order case for model A, we have $\theta(B) = 1 + \theta B$, which has root $B = -1/\theta$. Provided $|\theta| < 1$, the root $B = -1/\theta$ is real and lies **outside the unit circle**. So, model A is invertible if $|\theta| < 1$.

Conditions for Invertibility Using Backward Shift Operator

- 1 If we regard B as a complex variable, the operator $1/\theta(B)$ can be expanded as an infinite series, i.e.,

$$\frac{1}{1 + \theta B} = 1 + \sum_{i=1}^{\infty} (-\theta)^i B^i \quad (12)$$

- 2 When $|\theta| < 1$, this infinite series is convergent since

$$|1 + \sum_{i=1}^{\infty} (-\theta)^i| \leq 1 + \sum_{i=1}^{\infty} |\theta|^i = \frac{1}{1 - |\theta|} \quad (13)$$

- 3 Hence,

$$W_t = (1 + \sum_{i=1}^{\infty} (-\theta)^i B^i) X_t = X_t + \sum_{i=1}^{\infty} (-\theta)^i X_{t-i} \quad (14)$$

and thus $\{X_t\}$ is invertible.

- 4 We can use the same argument to prove invertibility of MA(q) process. Suppose that $\theta(B)$ can be decomposed as the following form

$$\theta(B) = (1 + \theta_1 B) \dots (1 + \theta_q B) \quad (15)$$

where $\theta_1, \dots, \theta_q$ could possibly take complex values. Then,

$$\frac{1}{\theta(B)} = \prod_{j=1}^q \frac{1}{1 + \theta_j B} = \prod_{j=1}^q (1 + \sum_{i=1}^{\infty} (-\theta_j)^i B^i). \quad (16)$$

Conditions for Invertibility Using Backward Shift Operator

- ① When all the roots, $-1/\theta_1, \dots, -1/\theta_q$, are outside the unit circle, the product of infinite series above is convergent, and hence $W_t = 1/\theta(B)X_t$ can be written in the form of infinite series and therefore $\{X_t\}$ is invertible.
- ② MA processes have been used in many areas, particularly econometrics. For example, economic indicators are affected by a variety of **random** events such as strikes, government decisions, shortages of key materials and so on. Such events will not only have an immediate effect but may also affect economic indicators to a lesser extent in several subsequent periods, and so it is at least plausible that an MA process may appropriate.
- ③ Note that an arbitrary constant, μ say, may be added to the right-hand side of the MA(q) equation as follows:

$$MA(q) : X_t = \mu + \beta_0 W_t + \beta_1 W_{t-1} + \dots + \beta_q W_{t-q} \quad (17)$$

where $\{\beta_i\}$ are constants, and W s are usually scaled so that $\beta_0 = 1$. It can then be shown that adding μ does not affect the autocorrelation coefficient function of the MA(q) process.

General-order Process

- 1 As in the first-order case, we can express an AR process of finite order as an MA process of infinite order. This may be done by successive substitution, or, more easily, by utilizing the backward shift operator. Thus equation of AR(p) process given by

$$AR(p) : X_t = \phi_1 X_{t-1} + \dots + \phi_p X_{t-p} + W_t \quad (18)$$

may be written as

$$(1 - \phi_1 B - \dots - \phi_p B^p) X_t = W_t \quad (19)$$

or equivalently as

$$X_t = \frac{W_t}{1 - \phi_1 B - \dots - \phi_p B^p} = f(B) W_t \quad (20)$$

where

$$f(B) = \frac{1}{1 - \phi_1 B - \dots - \phi_p B^p} = 1 + \beta_1 B + \beta_2 B^2 + \dots \quad (21)$$

- 2 The relationship between the ϕ s and the β s may then be found. Having expressed X_t as an MA process, it follows that $\mathbb{E}[X_t] = 0$. The variance is finite provided that $\sum \beta_i^2$ converges, and this is a necessary condition for **stationarity**.
- 3 The autocovariance function is given by

$$\gamma_k = \sigma_w^2 \sum_{i=0}^{\infty} \beta_i \beta_{i+k} \quad (22)$$

where $\beta_0 = 1$. A sufficient condition for this to converge, and hence for stationarity, is that $\sum |\beta_i|$ converges (these conditions are discussed in detail in the complementary slides).

Yule-Walker Equations

- ① We can in principle find the autocorrelation coefficient function for the general-order AR process using the procedure presented above, but the $\{\beta_i\}$ may be algebraically hard to find.
- ② The alternative simpler way is to assume the process is stationary, multiply through equation AR(p) by X_{t-k} , take expectations and divide by σ_X^2 , assuming that the variance of X_t is finite. Then, using the fact that $\rho_k = \rho_{-k}$ for all k , we find

$$\rho_k = \phi_1 \rho_{k-1} + \cdots + \phi_p \rho_{k-p} \quad \text{for all } k > 0. \quad (23)$$

- ③ This set of equations is called Yule-Walker equations and has the general solution

$$\rho_k = A_1 \pi_1^{|k|} + \cdots + A_p \pi_p^{|k|}, \quad (24)$$

where $\{\pi_i\}$ are the roots of the so-called **auxiliary equation**

$$y^p - \phi_1 y^{p-1} - \cdots - \phi_p = 0. \quad (25)$$

The constants $\{A_i\}$ are chosen to satisfy the initial conditions depending on $\rho_0 = 1$, which means that $\sum A_i = 1$.

- ④ The first $(p-1)$ Yule-Walker equations provide $(p-1)$ further restrictions on the $\{A_i\}$ using $\rho_0 = 1$ and $\rho_k = \rho_{-k}$.

Stationarity Conditions

- 1 From the general form of ρ_k , it is clear that ρ_k tends to zero as k increases provided that $|\pi_i| < 1$ for all i , and this is a **necessary** and **sufficient** condition for the AR(p) process to be **stationary**.
- 2 It can be shown that an equivalent way of expressing the stationarity condition is to say that the roots of the equation

$$\phi(B) = 1 - \phi_1 B - \dots - \phi_p B^p = 0 \quad (26)$$

must lie outside the unit circle, where we again regard B as a complex variable, rather than as an operator, so that the roots, which may be complex, are greater than one in modulus.

- 3 Let us consider the AR(2) process, where π_1 and π_2 are the roots of the quadratic equation

$$y^2 - \phi_1 y - \phi_2 = 0. \quad (27)$$

Here $|\pi_i| < 1$ if

$$\left| \frac{\phi_1 \pm \sqrt{\phi_1^2 + 4\phi_2}}{2} \right| < 1 \quad (28)$$

Stationarity Conditions

- 1 From this it can be shown that the stationarity region is the triangular region satisfying

$$\phi_1 + \phi_2 < 1 \quad (29)$$

$$\phi_1 - \phi_2 > -1 \quad (30)$$

$$\phi_2 > -1 \quad (31)$$

- 2 The roots are real if $\phi_1^2 + 4\phi_2 > 0$, in which case the autocorrelation function decreases exponentially with k .
- 3 The roots are complex if $\phi_1^2 + 4\phi_2 < 0$, in which case the autocorrelation function turns out to be a damped sinusoidal wave.
- 4 When the roots are real, we have

$$\rho_k = A_1 \pi_1^{|k|} + A_2 \pi_2^{|k|} \quad (32)$$

where the constants A_1, A_2 are also real and may be found as follows. Since $\rho_0 = 1$, we have

$$A_1 + A_2 = 1 \quad (33)$$

while the first Yule-Walker equation gives

$$\rho_1 = \phi_1 \rho_0 + \phi_2 \rho_{-1} = \phi_1 + \phi_2 \rho_1. \quad (34)$$

Stationarity Conditions

- 1 This equation may be solved to give

$$\rho_1 = \frac{\phi_1}{1 - \phi_2} \quad (35)$$

which in turn must equal

$$A_1\pi_1 + A_2\pi_2 = A_1\pi_1 + (1 - A_1)\pi_2. \quad (36)$$

Hence, we find

$$A_1 = \frac{\phi_1/(1 - \phi_2) - \pi_2}{\pi_1 - \pi_2} \quad (37)$$

$$A_2 = 1 - A_1 \quad (38)$$

and this enables us to write down the general form of the autocorrelation coefficient function of an AR(2) process with **real roots**.

- 2 As an example, consider an AR(2) process,

$$X_t = \frac{1}{3}X_{t-1} + \frac{2}{9}X_{t-2} + W_t \quad (39)$$

Note that the roots of

$$1 - \frac{1}{3}B - \frac{2}{9}B^2 = 0 \quad (40)$$

are -3 and $3/2$, where both have absolute values larger than 1. This indicates that X_t is stationary and its autocorrelation coefficient function can be calculated with the method above.

Stationarity Conditions

- 1 Consider the AR(2) process given by

$$X_t = X_{t-1} - \frac{1}{2}X_{t-2} + W_t. \quad (41)$$

Is this process stationary? If so, what is its autocorrelation coefficient function?

- 2 In order to answer the first question, we find the roots of

$$\phi(B) = 1 - B - \frac{1}{2}B^2 = 0. \quad (42)$$

The roots of this equation are complex, namely, $1 \pm i$. As the modulus of both roots exceeds one, the roots are both outside the unit circle and so the process is stationary.

- 3 In order to find autocorrelation function of the process, we use the first Yule-Walker equation to give

$$\rho_1 = \rho_0 - \frac{1}{2}\rho_{-1} = 1 - \frac{1}{2}\rho_1 \implies \rho_1 = \frac{2}{3}. \quad (43)$$

- 4 For $k \geq 2$, the Yule-Walker equations are

$$\rho_k = \rho_{k-1} - \frac{1}{2}\rho_{k-2}. \quad (44)$$

We could use these equations to find ρ_2 , then ρ_3 and so on by successive substitution.

A Note on AR Processes

- 1 AR processes have been applied to many situations in which it is reasonable to assume that the present value of a time series depends linearly on the immediate past values together with a random error.
- 2 The non-zero means may also be dealt with by rewriting the equation of AR(p) as follows:

$$X_t - \mu = \phi_1(X_{t-1} - \mu) + \cdots + \phi_p(X_{t-p} - \mu) + W_t \quad (45)$$

This does not affect the autocorrelation coefficient function.

- 3 You ought to be careful not to confuse stationary models with the causal ones. If the roots fall outside of unit circle in a complex plane and additionally W_t is independent from past observations $\{X_{t-1}, X_{t-2}, \dots\}$, then you can claim the your model is a stationary process. In lack of having the latter condition being satisfied or assumed, the model is causal but not necessarily a stationary one..
- 4 Your developed model being stationary, causal, stable, etc., does not imply that the true Data Generating Process adheres to the same properties as well. However, if you perform all modeling, estimation, diagnostics and analysis accurately, then developed model will get be powerful enough to capture the behavior of the underlying phenomenon delineated by the DGP.

Mixed ARMA Models

- 1 A useful class of models for time series is formed by combining MA and AR processes. A mixed autoregressive/moving-average process containing p AR terms and q MA terms is said to be an ARMA process of order (p, q) . It is given by

$$X_t = \phi_1 X_{t-1} + \cdots + \phi_p X_{t-p} + W_t + \beta_1 W_{t-1} + \cdots + \beta_q W_{t-q} \quad (46)$$

where $\{W_t\}$ is a purely random process with mean zero and variance σ_w^2 .

- 2 Using the backward shift operator B , this equation may be written in the form

$$\phi(B)X_t = \theta(B)W_t, \quad (47)$$

where $\phi(B), \theta(B)$ are polynomials of order p, q , respectively, such that

$$\phi(B) = 1 - \phi_1 B - \cdots - \phi_p B^p \quad (48)$$

and

$$\theta(B) = 1 + \beta_1 B + \cdots + \beta_q B^q \quad (49)$$

Mixed ARMA Models

- 1 The conditions on the model parameters to make the process **stationary** and **invertible** are the same as for a pure AR or pure MA process, namely, that the values of $\{\phi_i\}$, which make the process stationary, are such that the roots of

$$\phi(B) = 0 \quad (50)$$

lie outside the unit circle, while the values of $\{\beta_i\}$, which make the process invertible, are such that the roots of

$$\theta(B) = 0 \quad (51)$$

lie outside the unit circle.

- 2 The importance of ARMA processes lies in the fact that a stationary time series may often be adequately modeled by an ARMA model involving fewer parameters than a pure MA or AR process by itself. This is called the **Principle of Parsimony**, which states that we want to find a model with as few parameters as possible, but which gives an adequate representation of the data at hand

HW: Why ARMA has fewer parameters?

ARMA(1,1) Process

- 1 Consider the ARMA(1,1) process

$$X_t = \phi X_{t-1} + W_t + \beta W_{t-1} \quad (52)$$

where $|\phi| < 1$ and $|\beta| < 1$. Derive the autocorrelation coefficient function.

- 2 The Yule-Walker equations are

$$\rho_k = \phi \rho_{k-1} \quad k \geq 2 \quad (53)$$

To obtain the initial conditions, we note that γ_1 can be computed as follows:

$$\begin{aligned} \gamma_1 &= \text{Cov}(X_t, X_{t-1}) = \text{Cov}(\phi X_{t-1} + W_t + \beta W_{t-1}, X_{t-1}) \\ &= \phi \gamma_0 + \beta \text{Cov}(W_{t-1}, X_{t-1}) \\ &= \phi \gamma_0 + \beta \text{Cov}(W_{t-1}, \phi X_{t-2} + W_{t-1} + \beta W_{t-2}) \\ &= \phi \gamma_0 + \beta \sigma_w^2 \end{aligned}$$

ARMA(1,1) Process (Cont'd)

- 1 The variance of X_t , or γ_0 , can be calculated as follows:

$$\begin{aligned}\gamma_0 &= \text{Var}[\phi X_{t-1} + W_t + \beta W_{t-1}] \\ &= \text{Cov}(\phi X_{t-1} + W_t + \beta W_{t-1}, \phi X_{t-1} + W_t + \beta W_{t-1}) \\ &= \phi^2 \gamma_0 + (1 + \beta^2) \sigma_w^2 + 2\phi\beta \text{Cov}(X_{t-1}, W_{t-1}) \\ &= \phi^2 \gamma_0 + (1 + 2\phi\beta + \beta^2) \sigma_w^2 \\ &= \frac{1 + 2\phi\beta + \beta^2}{1 - \phi^2} \sigma_w^2\end{aligned}$$

- 2 Now, by plugging in the value of γ_0 , we compute γ_1 as follows:

$$\gamma_1 = \frac{(1 + \phi\beta)(\phi + \beta)}{1 - \phi^2} \sigma_w^2 \quad (56)$$

- 3 Then,

$$\rho_1 = \frac{\gamma_1}{\gamma_0} = \frac{(1 + \phi\beta)(\phi + \beta)}{1 + 2\phi\beta + \beta^2} \quad (57)$$

- 4 Using the Yule-Walker equations, we then have

$$\rho_k = \frac{(1 + \phi\beta)(\phi + \beta)}{1 + 2\phi\beta + \beta^2} \phi^{k-1} \quad k \geq 1. \quad (58)$$

Time Series Analysis

Exploiting the AR-MA Duality Relation

AR and MA Representations

- 1 It is sometimes helpful to express an ARMA model as a pure MA process in the form

$$X_t = \psi(B)W_t \quad (59)$$

where

$$\psi(B) = \sum_{i \geq 0} \psi_i B^i \quad (60)$$

is the MA operator, which may be of infinite order.

- 2 The ψ weights, $\{\psi_i\}$, can be useful in calculating forecasts and in assessing the properties of a model.
- 3 By comparison with

$$\phi(B)X_t = \theta(B)W_t \quad (61)$$

we see that

$$\psi(B) = \frac{\theta(B)}{\phi(B)}. \quad (62)$$

- 4 Alternatively, it can be helpful to express an ARMA model as a pure AR process in the form

$$\pi(B)X_t = W_t \quad (63)$$

where

$$\pi(B) = \frac{\phi(B)}{\theta(B)}. \quad (64)$$

AR and MA Representations

- 1 By convention we write

$$\pi(B) = 1 - \sum_{i>0} \pi_i B^i \quad (65)$$

since the natural way to write an AR model is in the form

$$X_t = \sum_{i=1}^{\infty} \pi_i X_{t-i} + W_t \quad (66)$$

- 2 By comparing Equations (62) and (64), we see that

$$\pi(B)\psi(B) = 1. \quad (67)$$

- 3 The ψ weights or π weights may be obtained directly by division or by equating powers of B in an equation such as

$$\psi(B)\phi(B) = \theta(B). \quad (68)$$

AR and MA Representations

- Find the ψ weights and π weights for the ARMA(1,1) process given by

$$X_t = 0.5X_{t-1} + W_t - 0.3W_{t-1}. \quad (69)$$

Here,

$$\phi(B) = 1 - 0.5B \quad (70)$$

and

$$\theta(B) = 1 - 0.3B. \quad (71)$$

It follows that the process is **stationary** and **invertible**, because both equations have roots greater than one (or are outside the unit circle). Then,

$$\psi(B) = \frac{\theta(B)}{\phi(B)} = \frac{1 - 0.3B}{1 - 0.5B} = (1 - 0.3B)(1 + 0.5B + 0.5^2B^2 + \dots) \quad (72)$$

which can be simplified as

$$\psi(B) = 1 + 0.2B + 0.1B^2 + 0.05B^3 + \dots \quad (73)$$

Hence,

$$\psi_0 = 1, \quad \psi_i = 0.2 \times 0.5^{i-1} \quad \text{for } i = 1, 2, \dots \quad (74)$$

Similarly, we find

$$\pi_0 = 1, \quad \pi_i = 0.2 \times 0.3^{i-1} \quad \text{for } i = 1, 2, \dots \quad (75)$$

- Note that both the ψ weights and π weights die away quickly, and this also indicates a stationary, invertible process.

HW: How do you interpret the statement above? In other words, what information do the ψ - and π -weights reveal?

Sample Problems

Exercise 4.7. Describe the important characteristics of the autocorrelation function for the following models:

MA(1): Has nonzero correlation only at lag 1. Could be positive or negative but must be between -0.5 and 0.5 (why?)

MA(2): Has nonzero correlation only at lags 1 and 2 (why?)

AR(1): Has exponentially decaying autocorrelations starting from lag 0. If $\phi > 0$, then all autocorrelations are positive. If $\phi < 0$, then autocorrelations alternate negative, positive negative, etc., (why?)

AR(2): Autocorrelations can have several patterns but if the roots of the characteristic equation are complex numbers, then the pattern will be a cosine with a decaying magnitude (why?)

ARMA(1,1): Has exponentially decaying autocorrelations starting from lag 1, but not from lag zero (why?)

Worked Out Example

Exercise 4.6. Suppose that $\{X_t\}$ is an AR(1) process with $-1 < \phi < 1$. Find the autocovariance function for $Y_t = \nabla X_t = X_t - X_{t-1}$ in terms of ϕ and σ_w^2 . Furthermore, derive the variance of the process.

Solution. Recall that for $k \geq 0$,

$$\text{Cov}(X_t, X_{t-k}) = \phi^k \frac{\sigma_w^2}{1 - \phi^2} \quad (76)$$

Then, for $k > 0$

$$\begin{aligned} \text{Cov}(Y_t, Y_{t-k}) &= \text{Cov}(X_t - X_{t-1}, X_{t-k} - X_{t-k-1}) \\ &= \text{Cov}(X_t, X_{t-k}) + \text{Cov}(X_t, -X_{t-k-1}) + \text{Cov}(-X_{t-1}, X_{t-k}) + \text{Cov}(-X_{t-1}, -X_{t-k-1}) \\ &= [\phi^k - \phi^{k+1} - \phi^{k-1} + \phi^k] \frac{\sigma_w^2}{1 - \phi^2} \\ &= [2\phi^k - \phi^{k+1} - \phi^{k-1}] \frac{\sigma_w^2}{1 - \phi^2} \\ &= [2\phi - \phi^2 - 1] \phi^k \frac{\sigma_w^2}{1 - \phi^2} = -[(1 - \phi)^2] \phi^{k-1} \frac{\sigma_w^2}{1 - \phi^2} = -\left[\frac{1 - \phi}{1 + \phi}\right] \phi^{k-1} \sigma_w^2 \end{aligned}$$

In particular,

$$\text{Var}[Y_t] = \text{Var}[X_t - X_{t-1}] = \text{Var}[X_t] + \text{Var}[X_{t-1}] - 2\text{Cov}(X_t, X_{t-1}) = 2(1 - \phi) \frac{\sigma_w^2}{1 - \phi^2} = \frac{2\sigma_w^2}{1 + \phi}$$

A Clarification on Infinite Notation in Time Series Representation

- ① We mentioned earlier that a time series could be thought as a specific realization of an infinite set of observations which we called it an ensemble.
- ② Now let us expand more on this concept to clarify the infinity notation which appears in many time series models.
- ③ A time series is basically a collection of observations indexed by the date of each observation. Usually we have collected data beginning at some particular data (say, $t=1$) and ending at another (say, $t = N$): $(x_1, x_2, \dots, x_N)$.
- ④ It is though often useful to imagine that we **could have** obtained earlier observations $(\dots, x_{-2}, x_{-1}, x_0)$ or later observations $(x_{N+1}, x_{N+2}, \dots)$, **had the process been observed for more time**.
- ⑤ The observed sample $(x_1, x_2, \dots, x_N)$ could then be viewed as a **finite segment** of a **doubly infinite sequence**, denoted by $\{x_t\}_{t=-\infty}^{\infty}$:

$$\{x_t\}_{t=-\infty}^{\infty} = \{\dots, x_{-1}, x_0, \underbrace{x_1, x_2, \dots, x_N}_{\text{observed sample}}, x_{N+1}, x_{N+2}, \dots\}$$

Linear Processes

- ④ The class of linear time series models, which includes the class of autoregressive moving-average (ARMA) models, provides a general framework for studying stationary processes.
- ⑤ In fact, every second-order stationary process (weak stationary process) is either a linear process or can be transformed to a linear process by subtracting a deterministic component. This result is known as Wold's decomposition theorem which we will get to it shortly.

Definition (Linear Process)

The stochastic process $\{X_t\}$ is a linear process if it has representation

$$X_t = \sum_{j=-\infty}^{\infty} \psi_j W_{t-j} \quad (78)$$

for all t , where $\{W_t\} \sim wn(0, \sigma_w^2)$ and $\{\psi_j\}$ is a sequence of constants with $\sum_{j=-\infty}^{\infty} |\psi_j| < \infty$.

In terms of the backward shift operator, B , this can be written more compactly as:

$$X_t = \psi(B)W_t \quad (79)$$

where $\psi(B) = \sum_{j=-\infty}^{\infty} \psi_j B^j$.

Linear Processes - Convergence Properties

A linear process is called a **moving average** or $MA(\infty)$ if

$$\psi_j = 0 \quad \text{for all } j < 0.$$

That is, if the process can be written as follows:

$$X_t = \mu + \sum_{j=0}^{\infty} \psi_j W_{t-j}. \quad (80)$$

Remark

The condition $\sum_{j=0}^{\infty} \psi_j < \infty$ ensures that:

1 $X_t = \sum_{j=-\infty}^{\infty} \psi_j W_{t-j}$ would converge, since $\mathbb{E}[|W_t|] \leq \sigma_w^2$ and

$$\mathbb{E}[|W_t|] \leq \sum_{j=-\infty}^{\infty} (|\psi_j| \mathbb{E}[|W_{t-j}|]) = \left(\sum_{j=-\infty}^{\infty} |\psi_j| \right) \sigma_w^2 < \infty \quad (81)$$

2 $\sum_{j=0}^{\infty} \psi_j^2 < \infty$ and hence that the series $X_t = \sum_{j=0}^{\infty} \psi_j W_{t-j}$ converges in **mean square**, i.e.,

$$\lim_{n \rightarrow \infty} \mathbb{E} \left[\left(X_t - \sum_{j=-n}^n \psi_j W_{t-j} \right)^2 \right] = 0 \quad (82)$$

The Infinite-Order Moving Average Processes

- 1 The $MA(q)$ and $MA(\infty)$ process can be written as follows

$$MA(q) : X_t = \mu + \sum_{j=0}^q \theta_j W_{t-j}, \quad MA(\infty) : X_t = \mu + \sum_{j=0}^{\infty} \psi_j W_{t-j} \quad (83)$$

with $\theta_0 = 1$ and $\{W_t\} \sim wn(0, \sigma_w^2)$. To preserve notational flexibility, we will use ψ s for the coefficients of an **infinite-order** moving average process and θ 's for the coefficients of a **finite-order** moving average process.

- 2 We will prove now in a more details the $MA(\infty)$ generates a well-defined covariance-stationary (weak stationary) process provided that $\sum_{j=0}^{\infty} \psi_j^2 < \infty$.
- 3 It is also often convenient to work with a slightly **stronger** condition $\sum_{j=0}^{\infty} |\psi_j| < \infty$.

Definition (Square Summable and Absolutely Summable Sequences)

A sequence of numbers $\{\psi_j\}_{j=0}^{\infty}$ satisfying $\sum_{j=0}^{\infty} \psi_j^2 < \infty$ is said to be **square summable**, whereas a sequence satisfying $\sum_{j=0}^{\infty} |\psi_j| < \infty$ is said to be **absolutely summable**.

Convergence Results for Infinite-Order Moving Average Processes

Absolute summability implies square-summability, but the converse does not hold in general. We prove this important result by the following lemma:

Lemma

The absolute summability of the moving average coefficients implies square-summability.

Proof.

Suppose that $\{\psi_j\}_{j=0}^{\infty}$ is absolutely summable. Then there exists an $N < \infty$ such that $|\psi_j| < 1$ for all $j \geq N$, implying that $\psi_j^2 < |\psi_j|$ for all $j \geq N$. Then,

$$\sum_{j=0}^{\infty} \psi_j^2 = \sum_{j=0}^{N-1} \psi_j^2 + \sum_{j=N}^{\infty} \psi_j^2 < \sum_{j=0}^{N-1} \psi_j^2 + \sum_{j=N}^{\infty} |\psi_j| \quad (84)$$

Also,

$$N < \infty \implies \sum_{j=0}^{N-1} \psi_j^2 < \infty \quad (85)$$

and

$$\{\psi_j\}_{j=0}^{\infty} \text{ is absolutely summable} \implies \sum_{j=0}^{\infty} |\psi_j| < \infty \quad (86)$$

Hence, $\sum_{j=0}^{\infty} \psi_j^2 < \infty$, establishing that absolute summability implies square-summability, which completes the proof. □

Square-summability and Mean Square Convergence of $MA(\infty)$

Now, we show that square-summability of the moving average coefficients implies that $MA(\infty)$ generates a *mean square convergent* random variable.

Lemma

The infinite series $MA(\infty)$ converges in mean square provided that $\sum_{j=0}^{\infty} \psi_j^2 < \infty$ is satisfied.

Proof.

To prove the lemma we resort to *Cauchy criterion*. Cauchy criterion states that $\sum_{j=0}^{\infty} a_j$ is convergent if and only if, for any $\epsilon > 0$, there exists a suitably large integer N such that, for any integer $M > N$,

$$\left| \sum_{j=0}^M a_j - \sum_{j=0}^N a_j \right| < \epsilon. \quad (87)$$

In other words, once we have summed N terms, calculating the sum out to a larger number M does not change the total by any more than an arbitrarily small number ϵ . For a stochastic process $MA(\infty)$, the Cauchy criterion states that $\sum_{j=0}^T \psi_j W_{t-j}$ converges if and only if, for any $\epsilon > 0$, there exist a suitably large integer N such that for any integer $M > N$:

$$\mathbb{E} \left[\left(\sum_{j=0}^M \psi_j W_{t-j} - \sum_{j=0}^N \psi_j W_{t-j} \right)^2 \right] < \epsilon \quad (88)$$

By simple computation, the left-hand-side of the inequality above, will reduce to $\left(\sum_{j=0}^M \psi_j^2 - \sum_{j=0}^N \psi_j^2 \right) \sigma_w^2$. But as required by the Lemma, $\sum_{j=0}^{\infty} \psi_j^2 < \infty$, so $\sum_{j=0}^{\infty} \psi_j^2$ converges, then by the Cauchy criterion the right-hand-side of the inequality above may be made as small as desired by choice of a suitably large N . This completes the proof. $\square$

Lemma

Let $\{Y_t\}$ be a *stationary* time series with mean 0 and covariance function γ' . If

$$\sum_{j=-\infty}^{\infty} |\psi_j| < \infty$$

then the time series

$$X_t = \sum_{j=-\infty}^{\infty} \psi_j Y_{t-j} = \psi(B)Y_t \quad (89)$$

is *stationary* with mean 0 and autocovariance function

$$\gamma_h = \sum_{j=-\infty}^{\infty} \sum_{k=-\infty}^{\infty} \psi_j \psi_k \gamma'_{h+k-j}. \quad (90)$$

In the special case where $\{X_t\}$ is a linear process,

$$\gamma_h = \sum_{j=-\infty}^{\infty} \psi_j \psi_{j+h} \sigma_w^2 \quad (91)$$

since if $\{Y_t\}$ is a white noise sequence $\{W_t\}$, then

$$\gamma'_{h-j+k} = \begin{cases} \sigma_w^2 & k = j - h \\ 0 & \text{o.w.} \end{cases} \quad (92)$$

Proof.

Based upon the stated remark, since $\sum_{j=-\infty}^{\infty} |\psi_j| < \infty$ then $X_t = \sum_{j=-\infty}^{\infty} \psi_j Y_{t-j}$ is guaranteed to converge. Then, since $\mathbb{E}[Y_t] = 0$,

$$\mathbb{E}[X_t] = \mathbb{E} \left[\sum_{j=-\infty}^{\infty} \psi_j Y_{t-j} \right] = \sum_{j=-\infty}^{\infty} \psi_j \mathbb{E}[Y_{t-j}] = 0 \quad (93)$$

and

$$\gamma_h = \mathbb{E} \left[\left(\sum_{j=-\infty}^{\infty} \psi_j Y_{t+h-j} \right) \left(\sum_{k=-\infty}^{\infty} \psi_k Y_{t-k} \right) \right] \quad (94a)$$

$$= \sum_{j=-\infty}^{\infty} \sum_{k=-\infty}^{\infty} \psi_j \psi_k \mathbb{E}[Y_{t+h-j} Y_{t-k}] = \sum_{j=-\infty}^{\infty} \sum_{k=-\infty}^{\infty} \psi_j \psi_k \gamma'_{h-j+k} \quad (94b)$$

which shows that $\{X_t\}$ is a **stationary** with autocovariance function

$$\gamma_h = \sum_{j=-\infty}^{\infty} \sum_{k=-\infty}^{\infty} \psi_j \psi_k \gamma'_{h+k-j}. \quad (95)$$

□

Remark

Note that the interchange of summation and expectation operations in the above calculations can be justified by the absolute summability of ψ_j and existence of second moment $\mathbb{E}[W_t^2]$. Specifically, if $\{X_t\}_{t=1}^{\infty}$ is a sequence of random variables such that $\sum_{t=1}^{\infty} \mathbb{E}|X_t| < \infty$, then

$$\mathbb{E} \left[\sum_{t=1}^{\infty} X_t \right] = \sum_{t=1}^{\infty} \mathbb{E}[X_t]. \quad (96)$$

More Comprehensive Analysis of AR(1)

- Earlier we discussed that AR(1) was defined as a **stationary** solution $\{X_t\}$ of the equation

$$AR(1) : X_t = \phi X_{t-1} + W_t \quad (97)$$

where $\{W_t\} \sim wn(0, \sigma_w^2)$, $|\phi| < 1$ and W_t uncorrelated with X_s for each $s < t$.

- However, the question remained to show that such a solution exists and more importantly it is a **unique stationary solution** of AR(1).
- To show existence of the solution, we consider the linear process defined by

$$X_t = \sum_{j=0}^{\infty} \phi^j W_{t-j} \quad (98)$$

Note that the coefficients ϕ^j are absolutely summable, since $|\phi| < 1$. It is then straightforward to verify that the process above is a solution of AR(1), and by preceding lemma, it is also stationary with mean 0 and autocovariance function

$$\gamma_h = \sum_{j=0}^{\infty} \phi^j \phi^{j+h} \sigma_w^2 = \frac{\sigma_w^2 \phi^h}{1 - \phi^2} \quad (99)$$

for lag parameter $h \geq 0$.

More Comprehensive Analysis of AR(1)

- ➊ In order to show

$$X_t = \sum_{j=0}^{\infty} \phi^j W_{t-j} \quad (100)$$

is the **only stationary** solution to AR(1) model, we previously derived a representation of X_t using recursive substitutions, i.e.,

$$X_t = \phi^k X_{t-k} + \sum_{j=0}^{k-1} \phi^j W_{t-j} \implies \phi^k X_{t-k} = X_t - \sum_{j=0}^{k-1} \phi^j W_{t-j}. \quad (101)$$

Using the fact from the stated lemma which implies $\sum_{j=0}^{k-1} \phi^j W_{t-j}$ is **mean square convergent**, we can write:

$$\lim_{k \rightarrow \infty} \mathbb{E} \left[\left(X_t - \sum_{j=0}^{k-1} \phi^j W_{t-j} \right)^2 \right] = \lim_{k \rightarrow \infty} \phi^{2k} \mathbb{E}[X_{t-k}^2] = 0 \quad (102)$$

- ➋ If $|\phi| < 1$ and variance of X_t is bounded. Hence, we can represent AR(1) as

$$X_t = \sum_{j=0}^{\infty} \phi^j W_{t-j} \quad (103)$$

in the mean square sense. Note that this is a linear process with $\psi_j = \phi^j$ for $j \geq 0$.

The Wold Decomposition

A general property of all members of the class of stationary stochastic processes pertains to a property that is known as Wold decomposition. The Wold decomposition exists for all stationary time series and provides a theoretical shortcut for obtaining properties of such processes.

Definition (Wold Decomposition)

If $\{X_t\}$ is a nondeterministic stationary time series, then $X_t = \mu_t + \sum_{j=0}^{\infty} \psi_j W_{t-j} + V_t$, where

- ① $\psi_0 = 1$
- ② $\sum_{j=0}^{\infty} \psi_j^2 < \infty$
- ③ $\{W_t\} \sim wn(0, \sigma_w^2)$
- ④ $Cov(W_s, V_t) = 0$ for all s and t
- ⑤ $\{V_t\}$ is deterministic

Remark

- ① *The quadratic convergence of the series of the $\{\psi_j\}$ guarantees the existence of second moments of the process.*
- ② *There is no need of any distributional assumption for this decomposition to hold.*
- ③ *There is no need of the W_t be independent, it is sufficient that they are uncorrelated.*

Deriving Properties of Stationary Time Series Using Wold Decomposition

- 1 Let us for now focus on **purely nondeterministic** processes in which all additive deterministic components, e.g. V_t , of the time series have been subtracted in advance. Note that by using its own lagged values, any deterministic component can be perfectly predicted in advance. This holds for a constant mean, periodical, polynomial or exponential series in t .

- 2 Thus, one can write

$$X_t = \mu_t + \sum_{j=0}^{\infty} \psi_j W_{t-j} \quad (104)$$

- 3 For the mean we get

$$\mathbb{E}[X_t] = \mu_t + \sum_{j=0}^{\infty} \psi_j \mathbb{E}[W_{t-j}] = \mu_t \quad (105)$$

- 4 For the variance we get

$$\text{Var}[X_t] = \sum_{j=0}^{\infty} \psi_j^2 \text{Var}[W_{t-j}] = \sigma_w^2 \sum_{j=0}^{\infty} \psi_j^2 \quad (106)$$

- 5 For the autocovariance we get

$$\gamma_h = \text{Cov}(X_t, X_{t+h}) = \sigma_w^2 \sum_{j=0}^{\infty} \psi_j \psi_{j+h} < \infty \quad (107)$$

Applications of Wold Decomposition

- 1 It then becomes clear that autocovariances are only functions of the time difference, i.e., the distance between two random variables. Thus all conditions of covariance stationarity (weak stationarity) are fulfilled.

- 2 Note that the autocorrelation coefficient function is given by

$$\rho_h = \frac{\sum_{j=0}^{\infty} \psi_j \psi_{j+h}}{\sum_{j=0}^{\infty} \psi_j^2}, \quad h = 1, 2, \dots \quad (108)$$

- 3 All stationary models discussed in our course can be represented on the basis of the Wold Decomposition. However, this representation is, above all, interesting for theoretical reasons: in practice, applications of models with an infinite number of parameters are hardly useful.

The Notion of Causality

An important concept in linear processes is that of **causality**. We say that $\{X_t\}$ is a causal function of $\{W_s, s \leq t\}$; i.e., the current status only relates to the past events, not the future. More formally,

Definition (Causality)

A linear process $\{X_t\}$ is *causal* (strictly, a causal function of $\{W_t\}$), if there is a

$$\psi(B) = \psi_0 + \psi_1 B + \psi_2 B^2 + \dots \quad (109)$$

with $\sum_{j=0}^{\infty} |\psi_j| < \infty$ such that

$$X_t = \psi(B)W_t = \sum_{j=0}^{\infty} \psi_j W_{t-j}. \quad (110)$$

Remark

In $AR(1)$, our primary interest to impose the condition $|\phi| < 1$, as discussed earlier, is actually in order to find a unique stationary solution $\{X_t\}$ which is **causal**. That is, X_t can be expressed in terms of the **current** and **past** values $W_s, s \leq t$.

For the noncausal case $|\phi| > 1$ in $AR(1)$, however, a less interesting situation occurs where X_t is expressed in terms of the future values of $W_s, s \geq t$. Even though there are some theoretical importance pertaining to noncausal models, in practice we are most interested in the causal models which deliver more realistic interpretations.

Lemma

$AR(1)$ process $\phi(B)X_t = W_t$ with $\phi(B) = 1 - \phi B$ is causal if and only if $|\phi| < 1$, or the root z_1 of the polynomial $\phi(z) = 1 - \phi z$ satisfies $|z_1| > 1$.

The Notion of Invertibility

Let us consider the $MA(1)$ model

$$X_t = W_t + \theta W_{t-1} \quad (111)$$

Obviously X_t is a causal function of $\{W_t\}$. But, we also encounter another important terminology here which is referred to as *invertibility*. Just as causality means that X_t is expressible in terms of $W_s, s \leq t$, invertibility means that W_t is expressible in terms of $\{X_s, s \leq t\}$.

Definition (Invertibility)

A linear process $\{X_t\}$ is *invertible* (strictly, a invertible function of $\{W_t\}$), if there is a

$$\pi(B) = \pi_0 + \pi_1 B + \pi_2 B^2 + \dots \quad (112)$$

with $\sum_{j=0}^{\infty} |\pi_j| < \infty$ such that

$$W_t = \pi(B)X_t = \sum_{j=0}^{\infty} \pi_j X_{t-j}. \quad (113)$$

Remark

*In $MA(1)$, our primary interest is to impose the condition $|\theta| < 1$ in order to find a unique stationary solution $\{W_t\}$ which is invertible. That is, W_t can be expressed in terms of the *current* and *past* values $X_s, s \leq t$.*

Lemma

$MA(1)$ process $X_t = \theta(B)W_t$ where $\theta(B) = 1 + \theta B$ is not invertible if and only if $|\theta| \geq 1$, or the root z_1 of the polynomial $\theta(z) = 1 + \theta z$ satisfies $|z_1| \leq 1$.

Causality for AR(p) Process

Our goal now is to find conditions on AR(p) so that the causality requirements are fulfilled for the stochastic process.

Definition

An AR(p) process $\{X_t\}$ is a stationary process that satisfies

$$X_t - \phi_1 X_{t-1} - \cdots - \phi_p X_{t-p} = W_t \quad (114)$$

where $\{W_t\} \sim wn(0, \sigma_w^2)$. Equivalently, we can say $\phi(B)X_t = W_t$, where

$$\phi(B) = 1 - \phi_1 B - \cdots - \phi_p B^p.$$

Lemma

A unique stationary solution to $\phi(B)X_t = W_t$ exists if and only if

$$\phi(z) = 1 - \phi_1 z - \cdots - \phi_p z^p = 0 \implies |z| \neq 1 \quad (115)$$

Further, this AR(p) process is causal if and only if

$$\phi(z) = 1 - \phi_1 z - \cdots - \phi_p z^p = 0 \implies |z| > 1 \quad (116)$$

Example

When the causality condition for AR(p) is satisfied, we can write

$$X_t = \psi(B)W_t \quad (117)$$

where $\psi(B) = \psi_0 + \psi_1 B + \psi_2 B^2 + \dots$ for some ψ_j s satisfying $\sum_{j=0}^{\infty} |\psi_j| < \infty$. The question is then how to calculate ψ_j s? One way is to match the coefficients:

$$\phi(B)X_t = W_t \quad \text{and} \quad X_t = \psi(B)W_t \quad (118)$$

$$\implies 1 = \psi(B)\phi(B)$$

$$\iff 1 = (\psi_0 + \psi_1 B + \psi_2 B^2 + \dots)(1 - \phi_1 B - \dots - \phi_p B^p)$$

$$\iff 1 = \psi_0,$$

$$0 = \psi_1 - \phi_1 \psi_0,$$

$$0 = \psi_2 - \phi_1 \psi_1 - \phi_2 \psi_0$$

$$\vdots$$

$$\iff \psi_0 = 1 \text{ and } \psi_j = 0 \ (j < 0) \text{ and } \phi(B)\psi_j = 0 \ (j > 0).$$

Invertibility for MA(q) Process

We turn now to the problem of investigating conditions of invertibility for an MA(q) process.

Definition

An MA(q) process $\{X_t\}$ is a stationary process that satisfies

$$X_t = W_t + \theta_1 W_{t-1} + \cdots + \theta_q W_{t-q} \quad (120)$$

where $\{W_t\} \sim wn(0, \sigma_w^2)$. Equivalently, we can say $X_t = \theta(B)W_t$ where

$$\theta(B) = 1 + \theta_1 B + \cdots + \theta_q B^q$$

.

Lemma

A unique stationary solution to $X_t = \theta(B)W_t$ exists if and only if

$$\pi(z) = 1 + \pi_1 z + \cdots + \pi_q z^q = 0 \implies |z| \neq 1 \quad (121)$$

Further, this MA(q) process is not invertible if and only if

$$\pi(z) = 1 + \pi_1 z + \cdots + \pi_q z^q = 0 \implies |z| \leq 1 \quad (122)$$

Example

When the invertibility condition for MA(q) is satisfied, we can write

$$W_t = \pi(B)X_t \quad (123)$$

where $\pi(B) = \pi_0 + \pi_1 B + \pi_2 B^2 + \dots$ for some π_j s satisfying $\sum_{j=0}^{\infty} |\pi_j| < \infty$. The next question again pertains to calculating π_j s? We may again use the method of matching the coefficients:

$$X_t = \theta(B)W_t \quad \text{and} \quad W_t = \pi(B)X_t \quad (124)$$

$$\implies 1 = \pi(B)\theta(B)$$

$$\iff 1 = (\pi_0 + \pi_1 B + \pi_2 B^2 + \dots)(1 + \theta_1 B + \dots + \theta_p B^p)$$

$$\iff 1 = \pi_0,$$

$$0 = \pi_1 + \pi_0 \theta_1,$$

$$0 = \pi_2 + \pi_1 \theta_1 + \pi_2 \theta_0$$

$$\vdots$$

$$\iff \pi_0 = 1 \text{ and } \pi_j = 0 \ (j < 0) \text{ and } \theta(B)\pi_j = 0 \ (j > 0).$$

ARMA Processes

We now consider an important class of linear processes known as *autoregressive moving average processes* (ARMA) which are defined by linear difference equations with constant coefficients.

Definition (ARMA(1,1))

The time series $\{X_t\}$ is an ARMA(1,1) process if it is stationary and satisfies for every t :

$$X_t - \phi X_{t-1} = W_t + \theta W_{t-1} \quad (126)$$

where $\{W_t\} \sim wn(0, \sigma_w^2)$ and $\phi + \theta \neq 0$. Using backward shift operator, the equation for ARMA(1,1) can be written as follows:

$$\phi(B)X_t = \theta(B)W_t \quad (127)$$

where

$$\phi(B) = 1 - \phi B \quad \text{and} \quad \theta(B) = 1 + \theta B. \quad (128)$$

Remark

Recall that we showed earlier that the linear process $X_t = \sum_{j=0}^{\infty} \psi_j W_{t-j}$ is absolutely convergent. This in turn implies that filters of the form

$$\alpha(B) = \sum_{j=-\infty}^{\infty} \alpha_j B^j \quad \text{and} \quad \beta(B) = \sum_{j=-\infty}^{\infty} \beta_j B^j \quad (129)$$

with absolutely summable coefficients can be applied successively to a stationary series $\{Y_t\}$ to generate a new stationary series

$$W_t = \sum_{j=-\infty}^{\infty} \psi_j Y_{t-j} \quad (130)$$

where

$$\psi_j = \sum_{k=-\infty}^{\infty} \alpha_k \beta_{j-k} = \sum_{k=-\infty}^{\infty} \beta_k \alpha_{j-k} \quad (131)$$

Remark

Using the previous remark and a bit of algebraic manipulation, we can conclude that the $MA(\infty)$ process

$$X_t = W_t + (\phi + \theta) \sum_{j=1}^{\infty} \phi^{j-1} W_{t-j} \quad (132)$$

is the unique solution to $ARMA(1,1)$.

- A stationary solution of the $ARMA(1,1)$ equations exists if and only if $\phi \neq \pm 1$.
- If $|\phi| < 1$, then unique stationary solution is given by the formula above. In this case, we say that $\{X_t\}$ is causal since X_t can be expressed in terms of the current and past values $W_s, s \leq t$.
- If $|\phi| > 1$, then the unique stationary solution is given by

$$X_t = -\theta\phi^{-1}W_t - (\theta + \phi) \sum_{j=1}^{\infty} \phi^{-j-1}W_{t+j} \quad (133)$$

This solution is noncausal, since X_t is then a function of $W_s, s \geq t$.

- If $|\theta| < 1$, then $ARMA(1,1)$ process is invertible, and W_t is expressed in terms of $X_s, s \leq t$ by

$$W_t = X_t - (\phi + \theta) \sum_{j=1}^{\infty} (-\theta)^{j-1} X_{t-j} \quad (134)$$

- If $|\theta| > 1$, then the $ARMA(1,1)$ process is noninvertible, and W_t is expressed in terms of $X_s, s \geq t$ by

$$W_t = -\phi\theta^{-1}X_t + (\theta + \phi) \sum_{j=1}^{\infty} (-\theta)^{-j-1} X_{t+j}. \quad (135)$$

Definition (ARMA(p,q))

$\{X_t\}$ is an ARMA(p,q) process if $\{X_t\}$ is stationary and if for every t ,

$$X_t - \phi_1 X_{t-1} - \dots - \phi_p X_{t-p} = W_t + \theta_1 W_{t-1} + \dots + \theta_q W_{t-q} \quad (136)$$

where $\{W_t\} \sim wn(0, \sigma_w^2)$ and the polynomials $(1 - \phi_1 z - \dots - \phi_p z^p)$ and $(1 + \theta_1 z + \dots + \theta_q z^q)$ have no common factors. Furthermore, the process $\{X_t\}$ is said to be an ARMA(p,q) process with mean μ if $\{X_t - \mu\}$ is an ARMA(p,q) process. It is also convenient to use the more concise form

$$\phi(B)X_t = \theta(B)W_t \quad (137)$$

where $\phi(\cdot)$ and $\theta(\cdot)$ are the p th and q th-degree polynomials

$$\phi(z) = 1 - \phi_1 z - \dots - \phi_p z^p \quad \text{and} \quad \theta(z) = 1 + \theta_1 z + \dots + \theta_q z^q \quad (138)$$

Lemma (Existence and Uniqueness)

A stationary solution $\{X_t\}$ of ARMA model exists (and is also the unique stationary solution) if and only if

$$\phi(z) = 1 - \phi_1 z - \dots - \phi_p z^p = 0 \implies |z| \neq 1. \quad (139)$$

Lemma (Causality)

An ARMA(p,q) process $\{X_t\}$ is causal, or a causal function of $\{W_t\}$, if there exist constants $\{\psi_j\}$ such that $\sum_{j=0}^{\infty} |\psi_j| < \infty$ and for all t

$$X_t = \sum_{j=0}^{\infty} \psi_j W_{t-j}. \quad (140)$$

Causality is equivalent to the condition

$$\phi(z) = 1 - \phi_1 z - \dots - \phi_p z^p = 0 \implies |z| > 1. \quad (141)$$

Remark

The sequence $\{\psi_j\}$ is determined by the relation $\psi(z) = \sum_{j=0}^{\infty} \psi_j z^j = \frac{\theta(z)}{\phi(z)}$, or by the identity

$$(1 - \phi_1 z - \dots - \phi_p z^p)(\psi_0 + \psi_1 z + \dots) = 1 + \theta_1 z + \dots + \theta_q z^q \quad (142)$$

Equating coefficients of z^j , $j = 0, 1, \dots$ we find that

$$\begin{aligned} 1 &= \psi_0 \\ \theta_1 &= \psi_1 - \psi_0 \phi_1 \\ \theta_2 &= \psi_2 - \psi_1 \phi_1 - \psi_0 \phi_2 \\ &\vdots \end{aligned}$$

or equivalently,

$$\psi_j - \sum_{k=1}^p \phi_k \psi_{j-k} = \theta_j, \quad j = 0, 1, 2, \dots \quad (144)$$

where

$$\theta_0 := 1 \quad \text{and} \quad \theta_j := 0 \text{ for } j > q \quad \text{and} \quad \psi_j := 0 \text{ for } j < 0. \quad (145)$$

Lemma (Invertibility)

An ARMA(p, q) process $\{X_t\}$ is invertible if there exist constants $\{\pi_j\}$ such that $\sum_{j=0}^{\infty} |\pi_j| < \infty$ and for all t

$$W_t = \sum_{j=0}^{\infty} \pi_j X_{t-j} \quad (146)$$

Invertibility is equivalent to the condition

$$\theta(z) = 1 + \theta_1 z + \cdots + \theta_q z^q \neq 0 \quad \text{for all } |z| \leq 1. \quad (147)$$

The sequence $\{\pi_j\}$ is also determined by the relation $\pi(z) = \sum_{j=0}^{\infty} \pi_j z^j = \frac{\phi(z)}{\theta(z)}$, or equivalently

$$\pi_j - \sum_{k=1}^q \theta_k \pi_{j-k} = -\phi_j, \quad j = 0, 1, 2, \dots \quad (148)$$

where

$$\phi_0 := 1 \quad \text{and} \quad \phi_j := 0 \text{ for } j > p \quad \text{and} \quad \pi_j := 0 \text{ for } j < 0. \quad (149)$$

Example (An ARMA(1,1) Process)

Consider the ARMA(1,1) process $\{X_t\}$ satisfying the equation

$$X_t - 0.5X_{t-1} = W_t + 0.4W_{t-1} \quad (150)$$

where $\{W_t\} \sim wn(0, \sigma_w^2)$.

Since the autoregressive polynomial $\phi(z) = 1 - 0.5z$ has a zero at $z = 2$, which is located outside the unit circle, we conclude that there exists a unique ARMA process satisfying the equation above that is also causal.

The coefficients $\{\psi_j\}$ in the $MA(\infty)$ representation of $\{X_t\}$ are found as follows:

$$\begin{aligned} \psi_0 &= 1 \\ \psi_1 &= 0.4 + 0.5 \\ \psi_2 &= 0.5(0.4 + 0.5) \\ \psi_j &= 0.5^{j-1}(0.4 + 0.5), \quad j = 1, 2, \dots \end{aligned}$$

The MA polynomial $\theta(z) = 1 + 0.4z$ has zero at $z = -1/0.4 = -2.5$, which is also located outside the unit circle. This implies that $\{X_t\}$ is invertible with coefficients $\{\pi_j\}$ given by

$$\begin{aligned} \pi_0 &= 1 \\ \pi_1 &= -(0.4 + 0.5) \\ \pi_2 &= -(0.4 + 0.5)(-0.4) \\ \pi_j &= -(0.4 + 0.5)(-0.4)^{j-1}, \quad j = 1, 2, \dots \end{aligned}$$

Example (An ARMA(2,1) Process)

Consider the ARMA(2,1) process defined by the equations

$$X_t - 0.75X_{t-1} + 0.5625X_{t-2} = W_t + 1.25W_{t-1} \quad (153)$$

where $\{W_t\} \sim wn(0, \sigma_w^2)$.

The AR polynomial $\phi(z) = 1 - 0.75z + 0.5625z^2$ has roots $z = 2(1 \pm i\sqrt{3})/3$, which lie outside the unit circle. The process is therefore causal. On the other hand, the MA polynomial $\theta(z) = 1 + 1.25z$ has zero at $z = -0.8$ and hence $\{X_t\}$ is not invertible.

Remark

It should be noted that causality and invertibility are not properties of the $\{X_t\}$ alone, but rather of the relationship between two processes $\{X_t\}$ and $\{W_t\}$ which appear in the defining equation of the ARMA model.

Remark

*The importance of ARMA processes lies in the fact that a stationary time series may often be adequately modelled by an ARMA model involving fewer parameters than a pure MA or AR process by itself. This is an example of what is often called the **Principle of Parsimony**. This says that we want to find a model with as few parameters as possible, but which gives an adequate representation of the data at hand.*